

Session 3 — How a telescope actually works

Practical Astronomy Intensive

9 questions 20 minutes 7 concepts 8 diagnostic

Classroom check with the instruments still on the table. The '600x!' box claim is attacked from three sides — aperture, magnification and the diffraction limit — because it is the misconception this session exists to kill, and it does not die to a single question.

1 A student has ₹40,000 and asks which is the best telescope design. What is the right answer?

telescope-designs

Apply

Multiple choice

- a There is no single best — refractor, reflector and catadioptric trade cost, size and image quality, and the best is the one you will actually carry outside ✓
- b The apochromatic refractor — it is the most expensive design, so it must be the better telescope
REVEALS Price taken as a proxy for capability; aperture-per-rupee ignored
- c Any of them, because optical design is the one specification that decides what a telescope shows
REVEALS Design treated as the deciding spec, displacing aperture
- d A refractor, because a reflector's secondary mirror and spider block the middle of the image and punch a hole in it
REVEALS Central obstruction believed to remove the middle of the image

At that budget a plain Dobsonian of twice the aperture will show more than a boutique apochromat, for less money. The secondary mirror costs a little light and contrast — every part of the target still reaches every part of the primary; it does not punch a hole. And the design that gets used is the design that wins.

2 Two telescopes, both at 100×: a 200 mm Dobsonian and a 100 mm refractor. What does the Dobsonian give you that the refractor cannot?

aperture

Apply

Multiple choice

- a Fainter stars and finer detail — four times the light grasp, since it climbs as the square of the diameter, and twice the resolving power ✓
- b Nothing, at the same magnification — the two are showing the same image, since magnification decides how much you can see
REVEALS Magnification believed to decide capability; aperture's role missed
- c Nothing that a dark enough site would not give the 100 mm refractor as well
REVEALS Belief that dark skies substitute for aperture on faint targets
- d A bigger image — the larger telescope's job is mainly to make things larger
REVEALS Bigger aperture believed mainly to enlarge, rather than to brighten and resolve

Aperture sets both the faintest star and the finest detail; power only enlarges what the aperture already delivered. Light grasp goes as diameter squared, so 200 mm collects 4× what 100 mm does. And below some aperture the photons are simply not there to be collected, however dark the site.

- 3 A 60 mm department-store refractor has '600× POWER!' printed in yellow across the box. In two or three sentences, explain to the shopper why that number is close to worthless — and say what number they should have looked at instead.

aperture

Reason

Short answer

ANSWER / WHAT TO WATCH FOR

Must say that magnification is the least important spec on the box, and that aperture — 60 mm here — is what decides both how faint you can go and how fine you can resolve. Should note that useful magnification is capped at roughly 50× per inch of aperture (about 120× for a 60 mm scope), so 600× delivers empty magnification: a dim, shaking, blurry image carrying no more information than the smaller one. Credit any answer that reaches for aperture, or light grasp, as the specification that matters.

This is the single misconception the session exists to kill, and it is worth making them say it out loud. The box quotes the one number that is free to inflate. Aperture costs money, so it is printed small.

- 4 Your Dobsonian has a focal length of 1200 mm. You fit a 6 mm eyepiece and then a 25 mm eyepiece. What powers do you get, and which will show more on Jupiter tonight?

magnification-eyepieces

Apply

Multiple choice

- a 200× and 48× — and which shows more depends on the night, since past the aperture's useful ceiling and on unsteady air the higher power just serves up a bigger blur ✓
- b 200× and 48× — the 6 mm always shows more detail, because more power always means more detail
REVEALS More power assumed always to mean more detail; empty magnification not understood
- c The powers cannot be worked out from this — magnification is a property of the telescope, and the eyepiece is a minor accessory
REVEALS Eyepiece treated as an accessory rather than half of the magnification
- d 7200× and 30000× — you multiply the two focal lengths
REVEALS Magnification formula not held: focal lengths multiplied rather than divided

Magnification = telescope focal length ÷ eyepiece focal length. The eyepiece is not an accessory — swapping eyepieces is the ONLY way to change power. Past roughly 50× per inch of aperture you get empty magnification, and on a boiling night the ceiling is set far lower still by the air.

- 5 The 60 mm box scope from the shop really does come with an eyepiece that yields 600×. If you fit it and point at Saturn, what will you actually see?

magnification-eyepieces

Reason

Multiple choice

- a Saturn much larger and with far more detail than at 60× — that is what the extra power is for
REVEALS The '600x!' box claim believed: magnification treated as the telescope's real power
- b A dim, shaking, badly blurred smear — 600× is far past what 60 mm can usefully deliver, so it enlarges without adding any information ✓
- c A large, sharp Saturn, though a dim one — magnification costs brightness but never detail
REVEALS Empty magnification not understood — power assumed to preserve detail while costing light
- d Nothing at all — the eyepiece is the wrong size for the scope and will not come to focus
REVEALS Eyepiece treated as a compatibility accessory rather than as half of the magnification

The eyepiece works exactly as advertised and gives 600×. That is the joke: 60 mm supports roughly 120× before the image goes empty, so the box's headline number is a promise the aperture cannot keep. Compute it, then look through it — that is what kills this one.

6 A student who does daytime photography says 'an f/5 telescope is faster, so stars will look brighter through it than through an f/10.' Is she right?

focal-ratio

Reason

Multiple choice

- a No — the f/10 is brighter, since a longer focal length collects light over a longer path
REVEALS f-ratio inverted, with focal length mistaken for a light-collecting quantity
- b Yes — a fast f-ratio gathers light more efficiently, so everything looks brighter through it
REVEALS Camera f-ratio intuition carried across to visual observing
- c Yes, and more generally f/5 is simply the better telescope — fast optics are better optics
REVEALS Fast f-ratio treated as straightforwardly better, rather than as a trade
- d No — visually, aperture and magnification set the brightness, and the f-ratio just follows from them; her camera intuition does not carry across to the eyepiece ✓

f-ratio = focal length ÷ aperture. It is not a free variable: visually, brightness comes from aperture and the magnification you are using. What f-ratio does buy is field — a fast scope frames nebulae widely at the price of fussier collimation and more demanding eyepieces, while slow optics suit planets.

7 A 200 mm Dobsonian at 20× gives an exit pupil of 10 mm. A student, aged 55, argues this is ideal because a bigger exit pupil always means more light into the eye. Where does the argument fail?

exit-pupil

Reason

Multiple choice

- a It fails because a 10 mm exit pupil causes blackouts and kidney-beaning, which is a fault in the eyepiece design
REVEALS Blackouts and kidney-beaning blamed on the eyepiece rather than on eye placement
- b His night pupil opens to about 5 mm, so half the beam lands on his iris and is thrown away — the extra aperture is wasted at that power ✓
- c It does not fail — a bigger exit pupil is always better, since more of the beam reaches the eye
REVEALS Bigger exit pupil assumed always better; the eye's pupil ceiling ignored
- d It fails only because he has not accounted for eye relief, which is purely a comfort matter anyway
REVEALS Eye relief dismissed as comfort rather than a limit on seeing the full field

Exit pupil = aperture ÷ magnification. Past what the night pupil can open to — about 7 mm young, 5 mm older — the surplus lands on the iris and is discarded. This is exactly why 7×50s (7 mm exit pupil) suit youngsters and 8×32s (4 mm) suit elders. Eye relief is a separate matter and not merely comfort: with 8 mm of it, a spectacle wearer physically cannot take in the whole field.

- 8 A tight double star will not split in a 60 mm refractor. A student swaps to a shorter eyepiece, then adds a Barlow, and is baffled that the blur only gets larger. Why?

resolution-limit

Reason

Multiple choice

- a He has not gone far enough — with enough magnification any double will eventually split
REVEALS Belief that enough magnification eventually reveals more detail; the diffraction limit not accepted
- b The tiny disc and rings around each star are a flaw in that cheap objective — a better 60 mm would resolve it
REVEALS Airy disc and rings mistaken for an optical defect rather than diffraction
- c The refractor's focal length is too short — a longer 60 mm scope would resolve finer
REVEALS Resolution attributed to focal length rather than to aperture alone
- d Diffraction sets a hard floor — the Airy discs of 60 mm simply overlap at that separation, and power only enlarges the overlapping blur ✓

Every perfect telescope turns a point source into an Airy disc; the aperture alone decides how small (Dawes: about $116/D$ arcsec, D in mm). Below that separation there is nothing more to reveal, and magnification just enlarges the disc and its rings. Splitting tight doubles needs inches, not power.

- 9 A cold front has just passed. The night is crystal-clear, sparkling, with the stars twinkling hard and a brisk wind. A student declares it perfect for Jupiter. What should you tell him?

seeing-transparency

Reason

Multiple choice

- a He is right, and he should carry the scope straight out and observe immediately to make the most of it
REVEALS Scope assumed ready the moment it goes outside — cool-down and tube currents ignored
- b Transparency is excellent but the seeing is probably terrible — that hard twinkle is the same air boiling Jupiter's disc into mush; a hazy still night usually beats it for planets ✓
- c He is right — a clear night with no haze is by definition good for everything, planets included
REVEALS Seeing confused with transparency — clarity taken as the single measure of a good night
- d He is right about the sky, and if Jupiter still shimmers into mush the telescope's optics are the problem
REVEALS Poor planetary image blamed on optics rather than on the atmosphere

Two independent things. Transparency is how faint you can go; seeing is how steady the air is. The freshly-scrubbed post-front night is the classic case: superb for faint galaxies, dreadful for planets. No optics can out-resolve the air. And a warm mirror carried straight outside makes its own tube currents, wrecking the view just as thoroughly.